Lecture 19 - Counting Scribe Notes

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1 Identical Variables

In the cases with identical variables, there are two further options for possible for determining possible distributions: no required minimum and required minimum or maximum.

1.1 No Requirements

This a case where there are no minimum requirements for the distribution, meaning that there can be possible distributions where someone has 0 of an item.

1.1.1 Example 1

There are 8 identical cookies which must be distributed among 3 people: A, B, C. So

$$X_a + X_b + X_c = 8 \tag{1}$$

where X_a, X_b, and X_c are the cookies each person receives. The number of possible ways these cookies can be split can be found by dividing the group of cookies using 2 bars. One such configuration is: XX X — X X — X X X. The cookies in this example can be represented as n, while the number of people can be represented by k. This gives the general relation of

$$\binom{n+k-1}{k-1} \tag{2}$$

1.2 Minimum Requirement

Another scenario which may be given for duplicate allowed cases are similar to the previous example with additional minimum requirements.

1.2.1 Example 2

There are 12 identical cookies and 4 people, however each person must get at least 1 cookie. In this example n is 12 and k is 4. Since each person needs to get at least one cookie, we can first assign each of the four people one cookie. Following this simplification, n is 8 and k is still 4, and using the provided equation

$$\binom{8+4-1}{4-1} = \binom{11}{3} \tag{3}$$

It is important to note that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \tag{4}$$

1.2.2 Example 3

It is also possible to have minimum requirements which are greater than one and specific to each variable. For example: there are 12 identical cookies and 3 people A, B, C. Where A must get at least 2 cookies, B at least 3 and C at least 1. This may be solved in the same manner as example 2, where each person is assigned there required minimum and n is reduced to the remaining cookies such that the equation becomes:

$$\binom{6+3-1}{3-1} = \binom{8}{2} \tag{5}$$

1.3 Maximum Requirement

For cases with maximum distribution requirements, the problem can often be simplified with the use of the complementing scenario. For example, if we want to know the number of ways to distribute 5 cookies between 3 people, but person A can only have a maximum of 2, then we can find the number of distributions where A receives more than 2 and subtract from 1.

1.3.1 Example 4

There are 8 cookies and 3 people: A, B, C. However, A can only have 3 cookies or less. In this case, the complement would be that A gets more than 3 cookies, or that A gets at least 4 cookies. The number of distribution for the complement can be found in the same manner as in the examples 2 and 3, resulting in

$$\binom{4+4-1}{3-1} = \binom{7}{2} \tag{6}$$

To find the actual number of distributions this can simply be subtracted from 1.

1.3.2 Example 5

There are 8 cookies and 3 people: A, B, C. Where no person can receive more than 4 cookies. Since each person has a max limit unlike in the previous example, the solution is slightly different. To simplify the calculation of the number of distribution we will first find the number of possible distribution as if there is no maximum limit. This results in the following:

$$\binom{8+3-1}{3-1} = \binom{10}{2} \tag{7}$$

For this specific case, it is not possible for two people to both receive more than 4 cookies at the same time. This allows us to simply fix the amount of cookies to at least 5 for any one person and calculate the number of possible distributions again. This time the number of cookies, n, is 3.

$$\binom{3+3-1}{3-1} = \binom{5}{2} \tag{8}$$

And to get the final solution, we subtract the answer of equation 8 from the answer of equation 7. If it was possible for two people to have more than 4 cookies as the same time, then we would need to compute another round of possible distributions for the case where two people are fixed to get at least 5 cookies. This would again need to be subtracted from the total number of possible distributions (equation 7).

1.4 Special Criteria- Example 6

There are 12 identical cookies and 4 people, but the cookies have to be distributed in pairs. In this case, the problem can be reduced to 6 pairs of cookies and 4 people ans solved as in Example 1 to get

$$\binom{6+4-1}{4-1} = \binom{9}{3} \tag{9}$$

2 Unique Variables

The other scenario in determining possible permutations is when the variable being distributed are unique, or non-identical.

2.1 Example 7

There are 6 different snacks and 3 people: A, B, C. If each person can pick the same snack then the possible combinations are

$$6 * 6 * 6 = 216 \tag{10}$$

We get this result because each person can pick one of the six snacks, or that there are six different possibilities for each person. Then for three people we must multiply them together to see the total number of possible distributions. However, if we consider distributing 1 snack to each person

$$3 * 3 * 3... = 3^6 = 729 \tag{11}$$

This result uses the same logic, where instead of focusing on the people we now focus on the snacks' options. Since, each snack has three options and there are six snacks we end up with the result above.

2.2 Example 8

Modifying example 6, where instead of having identical cookies we now have unique cookies and 4 people, with cookies having to be distributed in pairs. Each pair can have different cookies. We again make the simplification by reducing the number of cookies by placing them in pairs. However, in this case we are assuming only one of the possible pairing for each cookie as they are not identical. Following the last example, we can determine the amount of possible allocations for this set of pairings by

$$4 * 4 * 4... = 4^6 = 4096 \tag{12}$$

Then to determine the number of possible pairings of cookies, we use the basic property of n choose k, where n is number of remaining cookies and k represents the 2 cookies we choose to be in a pair. This gives us the following

$$\binom{12}{2}\binom{10}{2}\binom{8}{2}\binom{6}{2}\binom{4}{2}\binom{2}{2} \tag{13}$$

We get this result by choosing one pair from the 12 total cookies, then another from the remaining 10, then another pair and so forth until we are out of cookies. The total number of possible distributions is then the sum of the possible allocations for each pair times the number of pairs. However, this would only be the final answer if we were not concerned about duplicates. If we don't want duplicates the sum must be divided by 6!.

3 Introduction to Pascal's Triangle

Pascal's triangle states the following relation:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k} \tag{14}$$

This relation expands the ways to pick k number of elements from a group of n elements by reducing it into two parts. The first part represents selecting an item 'a' where the number of ways to pick 'a' are

$$\binom{n-1}{k-1} \tag{15}$$

And the number of ways that don't pick 'a' are

$$\binom{n-1}{k} \tag{16}$$

This relation holds true because it accounts for both possibilities for a fixed item 'a,' where it is selected and where it isn't. For the case when 'a' is picked it reduces the number of possible options and the number of items needed to be picked. And in the case when it's not selected, it can be excluded from the total number of item.